Observation of Gravitational Waves from a Binary Black Hole Merger

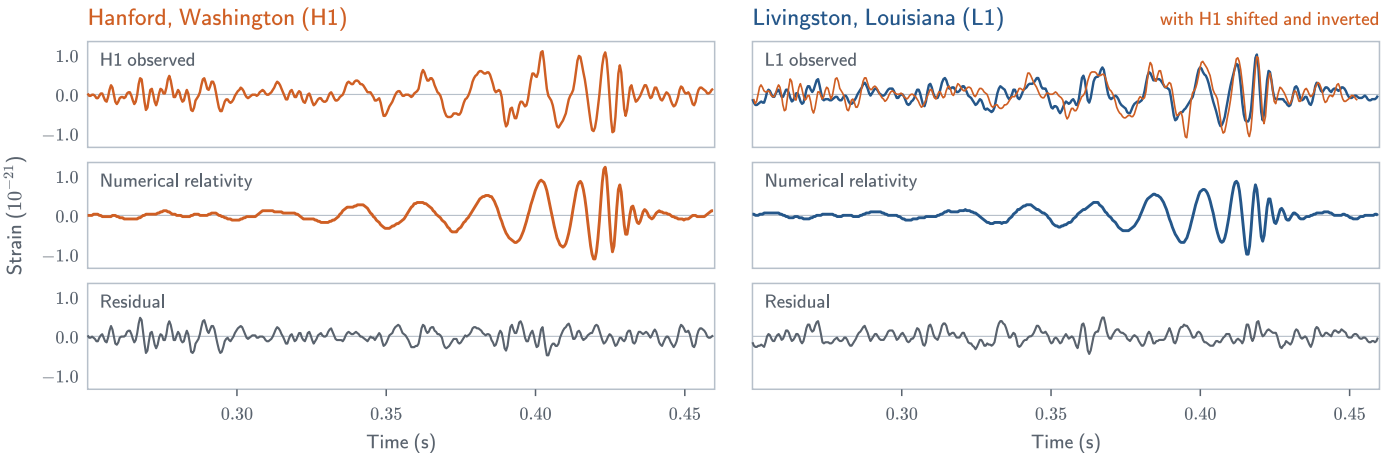


FIG. 1. The gravitational-wave event GW150914 observed by the LIGO Hanford (H1, left column panels) and Livingston (L1, right column panels) detectors. Times are shown relative to September 14, 2015 at 09:50:45 UTC. For visualization, all time series are filtered with a 35–350 Hz bandpass filter to suppress large fluctuations outside the detectors' most sensitive frequency band, and band-reject filters to remove the strong instrumental spectral lines. Top row, left: H1 strain. Top row, right: L1 strain. GW150914 arrived first at L1 and $6.9^{+0.5}_{-0.4}$ ms later at H1; for a visual comparison, the H1 data are also shown, shifted in time by this amount and inverted (to account for the detectors' relative orientations). Second row: Gravitational-wave strain projected onto each detector in the 35–350 Hz band. Solid lines show a numerical relativity waveform for a system with parameters consistent with those recovered from GW150914 confirmed to 99.9% by an independent calculation. Third row: Residuals after subtracting the filtered numerical relativity waveform from the filtered detector time series.

Redrawn from GWOSC open data (LIGO and Virgo Collaborations, CC BY 4.0); reconstructions and time–frequency row omitted.

B. P. Abbott *et al.*[a]
(LIGO Scientific Collaboration and Virgo Collaboration)

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On September 14, 2015 at 09:50:45 UTC the two detectors of the Laser Interferometer Gravitational-Wave Observatory simultaneously observed a transient gravitational-wave signal. The signal sweeps upwards in frequency from 35 to 250 Hz with a peak gravitational-wave strain of 1.0×10^{-21}. It matches the waveform predicted by general relativity for the inspiral and merger of a pair of black holes and the ringdown of the resulting single black hole. The signal was observed with a matched-filter signal-to-noise ratio of 24 and a false alarm rate estimated to be less than 1 event per 203 000 years, equivalent to a significance greater than 5.1σ. The source lies at a luminosity distance of 410^{+160}_{-180} Mpc corresponding to a redshift $z = 0.09^{+0.03}_{-0.04}$. In the source frame, the initial black hole masses are $36^{+5}_{-4}\,M_\odot$ and $29^{+4}_{-4}\,M_\odot$, and the final black hole mass is $62^{+4}_{-4}\,M_\odot$, with $3.0^{+0.5}_{-0.5}\,M_\odot c^2$ radiated in gravitational waves. All uncertainties define 90% credible intervals. These observations demonstrate the existence of binary stellar-mass black hole systems. This is the first direct detection of gravitational waves and the first observation of a binary black hole merger.

I. Introduction

In 1916, the year after the final formulation of the field equations of general relativity, Albert Einstein predicted the existence of gravitational waves. He found that the linearized weak-field equations had wave solutions: transverse waves of spatial strain that travel at the speed of light, generated by time variations of the mass quadrupole moment of the source [1,2]. Einstein understood that gravitational-wave amplitudes would be remarkably small; moreover, until the Chapel Hill

[a] Full author list given at the end of the published article; this abridged re-setting does not reproduce it.

conference in 1957 there was significant debate about the physical reality of gravitational waves [3].

The discovery of the binary pulsar system PSR B1913+16 by Hulse and Taylor [4] and subsequent observations of its energy loss by Taylor and Weisberg [5] demonstrated the existence of gravitational waves. This discovery, along with emerging astrophysical understanding [6], led to the recognition that direct observations of the amplitude and phase of gravitational waves would enable studies of additional relativistic systems and provide new tests of general relativity, especially in the dynamic strong-field regime.

A century after the fundamental predictions of Einstein and Schwarzschild, we report the first direct detection of gravitational waves and the first direct observation of a binary black hole system merging to form a single black hole. Our observations provide unique access to the properties of space-time in the strong-field, high-velocity regime and confirm predictions of general relativity for the nonlinear dynamics of highly disturbed black holes.[b]

II. Observation

On September 14, 2015 at 09:50:45 UTC, the LIGO Hanford, WA, and Livingston, LA, observatories detected the coincident signal GW150914 shown in Fig. 1. The initial detection was made by low-latency searches for generic gravitational-wave transients [7] and was reported within three minutes of data acquisition [8]. Subsequently, matched-filter analyses that use relativistic models of compact binary waveforms [9] recovered GW150914 as the most significant event from each detector for the observations reported here. Occurring within the 10-ms intersite propagation time, the events have a combined signal-to-noise ratio (SNR) of 24 [10].

Only the LIGO detectors were observing at the time of GW150914. The Virgo detector was being upgraded, and GEO 600, though not sufficiently sensitive to detect this event, was operating but not in observational mode. With only two detectors the source position is primarily determined by the relative arrival time and localized to an area of approximately 600 deg^2 (90% credible region) [11,12].

The basic features of GW150914 point to it being produced by the coalescence of two black holes—i.e., their orbital inspiral and merger, and subsequent final black hole ringdown. Over 0.2 s, the signal increases in frequency and amplitude in about 8 cycles from 35 to 150 Hz, where the amplitude reaches a maximum. The most plausible explanation for this evolution is the inspiral of two orbiting masses, m_1 and m_2, due to gravitational-wave emission. At the lower frequencies, such evolution is characterized by the chirp mass [13]

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} = \frac{c^3}{G}\left[\frac{5}{96}\,\pi^{-8/3} f^{-11/3}\,\dot{f}\right]^{3/5}, \quad (1)$$

where f and $\dot{f}$ are the observed frequency and its time derivative and G and c are the gravitational constant and speed of light. Estimating f and $\dot{f}$ from the data in Fig. 1, we obtain a chirp mass of $\mathcal{M} \simeq 30\,M_\odot$, implying that the total mass $M = m_1 + m_2$ is $\gtrsim 70\,M_\odot$ in the detector frame. This bounds the sum of the Schwarzschild radii of the binary components to $2GM/c^2 \gtrsim 210$ km. To reach an orbital frequency of 75 Hz (half the gravitational-wave frequency) the objects must have been very close and very compact; equal Newtonian point masses orbiting at this frequency would be only $\simeq 350$ km apart. A pair of neutron stars, while compact, would not have the required mass, while a black hole neutron star binary with the deduced chirp mass would have a very large total mass, and would thus merge at much lower frequency. This leaves black holes as the only known objects compact enough to reach an orbital frequency of 75 Hz without contact. Furthermore, the decay of the waveform after it peaks is consistent with the damped oscillations of a black hole relaxing to a final stationary Kerr configuration. Below, we present a general-relativistic analysis of GW150914; Fig. 2 shows the calculated waveform using the resulting source parameters.

III. Detectors

Gravitational-wave astronomy exploits multiple, widely separated detectors to distinguish gravitational waves from local instrumental and environmental noise, to provide source sky localization, and to measure wave polarizations. The LIGO sites each operate a single Advanced LIGO detector [14], a modified Michelson interferometer (see Fig. 3) that measures gravitational-wave strain as a difference in length of its orthogonal arms. Each arm is formed by two mirrors, acting as test masses, separated by $L_x = L_y = L = 4$ km. A passing gravitational wave effectively alters the arm lengths such that the measured difference is $\Delta L(t) = \delta L_x - \delta L_y = h(t)L$, where h is the gravitational-wave strain amplitude projected onto the detector. This differential length variation alters the phase difference between the two light fields returning to the beam splitter, transmitting an optical signal proportional to the gravitational-wave strain to the output photodetector.

To achieve sufficient sensitivity to measure gravitational waves, the detectors include several enhancements to the basic Michelson interferometer. First, each arm contains a resonant optical cavity, formed by its two test mass mirrors, that multiplies the effect of a gravitational wave on the light phase by a factor of 300 [15]. Second, a partially transmissive power-recycling mirror at the input provides additional resonant buildup of the laser light in the interferometer as a whole [16,17]: 20 W of laser input is increased to 700 W incident on the beam splitter, which is further increased to 100 kW circulating in each arm cavity. Third, a partially transmissive signal-recycling mirror at the output optimizes the gravitational-wave signal extraction by broadening the bandwidth of the arm cavities [18,19].

[b] Abridged here: Sections IV (Detector validation) and VII (Outlook), parts of Sections I, III, V and VI, some figure panels and most acknowledgments are left out; sections and references are renumbered.

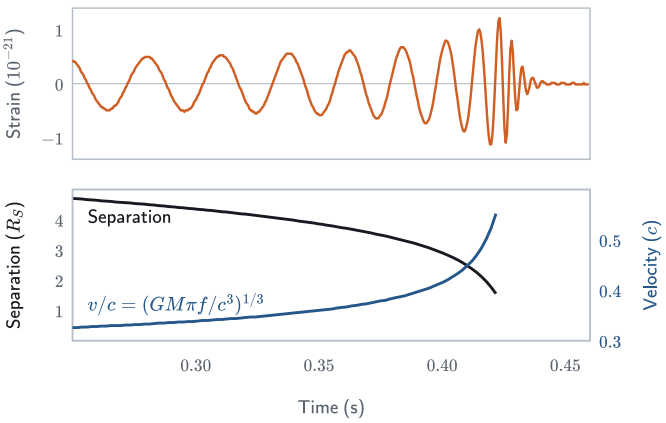


FIG. 2. Top: Estimated gravitational-wave strain amplitude from GW150914 projected onto H1. This shows the full bandwidth of the waveforms, without the filtering used for Fig. 1. Bottom: The Keplerian effective black hole separation in units of Schwarzschild radii ($R_S = 2GM/c^2$) and the effective relative velocity given by the post-Newtonian parameter $v/c = (GM\pi f/c^3)^{1/3}$, where f is the gravitational-wave frequency calculated with numerical relativity and M is the total mass (value from Table I).

Redrawn from GWOSC data (CC BY 4.0); insets omitted.

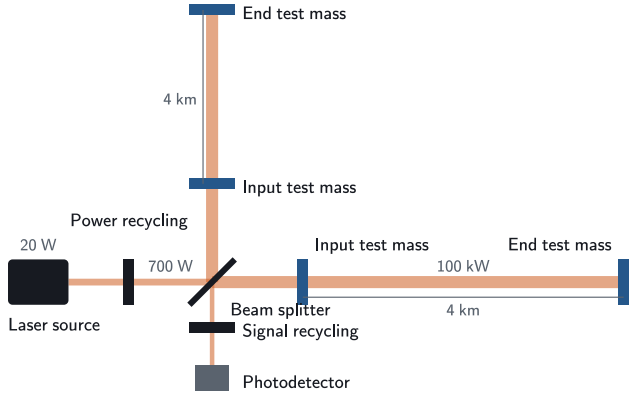


FIG. 3. Simplified diagram of an Advanced LIGO detector (not to scale). A gravitational wave propagating orthogonally to the detector plane and linearly polarized parallel to the 4-km optical cavities will have the effect of lengthening one 4-km arm and shortening the other during one half-cycle of the wave; these length changes are reversed during the other half-cycle.

Redrawn after the published diagram; its insets (a) and (b) are omitted.

IV. Searches

We present the analysis of 16 days of coincident observations between the two LIGO detectors from September 12 to October 20, 2015. This is a subset of the data from Advanced LIGO's first observational period that ended on January 12, 2016.

GW150914 is confidently detected by two different types of searches. One aims to recover signals from the coalescence of compact objects, using optimal matched filtering with waveforms predicted by general relativity. The other search targets a broad range of generic transient signals, with minimal assumptions about waveforms. These searches use independent methods, and their response to detector noise consists of different, uncorrelated, events. However, strong signals from binary black hole mergers are expected to be detected by both searches.

A. Generic transient search

Designed to operate without a specific waveform model, this search identifies coincident excess power in time-frequency representations of the detector strain data [8,20], for signal frequencies up to 1 kHz and durations up to a few seconds.

The search reconstructs signal waveforms consistent with a common gravitational-wave signal in both detectors using a multidetector maximum likelihood method. Each event is ranked according to the detection statistic

$$\eta_c = \sqrt{2E_c/(1 + E_n/E_c)},$$

where E_c is the dimensionless coherent signal energy obtained by cross-correlating the two reconstructed waveforms, and E_n is the dimensionless residual noise energy after the reconstructed signal is subtracted from the data. The statistic η_c thus quantifies the SNR of the event and the consistency of the data between the two detectors.

Based on their time-frequency morphology, the events are divided into three mutually exclusive search classes, as described in [7]: events with time-frequency morphology of known populations of noise transients (class C1), events with frequency that increases with time (class C3), and all remaining events (class C2).

Detected with $\eta_c = 20.0$, GW150914 is the strongest event of the entire search. Consistent with its coalescence signal signature, it is found in the search class C3 of events with increasing time-frequency evolution. Measured on a background equivalent to over 67 400 years of data and including a trials factor of 3 to account for the search classes, its false alarm rate is lower than 1 in 22 500 years. This corresponds to a probability $< 2 \times 10^{-6}$ of observing one or more noise events as strong as GW150914 during the analysis time, equivalent to 4.6σ. The left panel of Fig. 4 shows the C3 class results and background.

B. Binary coalescence search

This search targets gravitational-wave emission from binary systems with individual masses from 1 to 99 $M_\odot$, total mass less than 100 $M_\odot$, and dimensionless spins up to 0.99 [9]. To model systems with total mass larger than 4 $M_\odot$, we use the effective-one-body formalism [21], which combines results from the post-Newtonian approach [13,22] with results from black hole perturbation theory and numerical relativity. The waveform model [23,24] assumes that the spins of the merging objects are aligned with the orbital angular momentum, but the resulting templates can, nonetheless, effectively recover systems with misaligned spins in the parameter region of

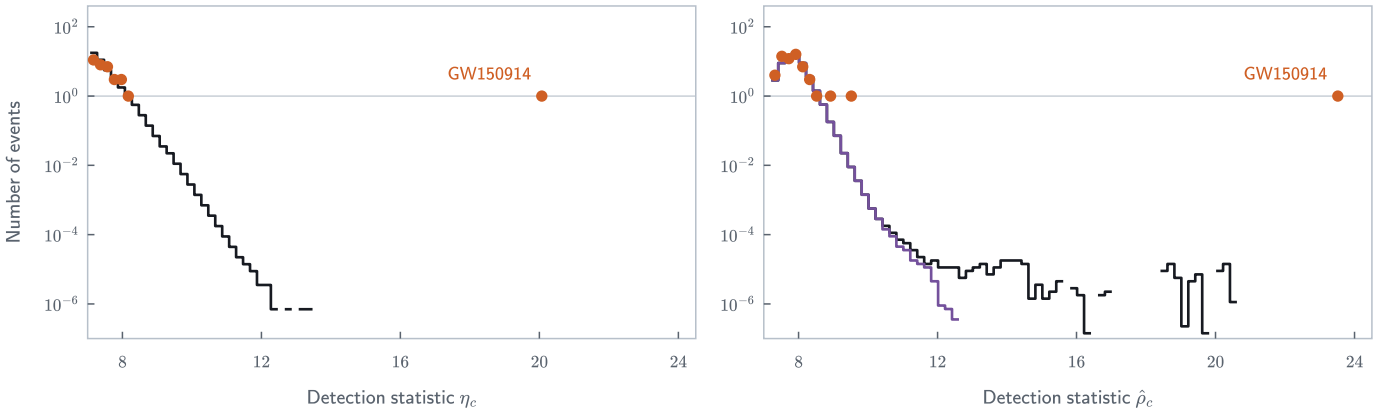


FIG. 4. Search results from the generic transient search (left) and the binary coalescence search (right). These histograms show the number of candidate events (orange markers) and the mean number of background events (black lines) in the search class where GW150914 was found as a function of the search detection statistic and with a bin width of 0.2. The significance of GW150914 is greater than 5.1σ and 4.6σ for the binary coalescence and the generic transient searches, respectively. Right: The tail in the black-line background of the binary coalescence search is due to random coincidences of GW150914 in one detector with noise in the other detector. (This type of event is practically absent in the generic transient search background because they do not pass the time–frequency consistency requirements used in that search.) The purple curve is the background excluding those coincidences, which is used to assess the significance of the second strongest event.

Redrawn from GWOSC data (CC BY 4.0); the C2 + C3 search and the significance scales omitted.

GW150914 [9]. Approximately 250 000 template waveforms are used to cover this parameter space.

The search calculates the matched-filter signal-to-noise ratio $\rho(t)$ for each template in each detector and identifies maxima of $\rho(t)$ with respect to the time of arrival of the signal [25–27]. For each maximum we calculate a chi-squared statistic χ_r^2 to test whether the data in several different frequency bands are consistent with the matching template [28]. Values of χ_r^2 near unity indicate that the signal is consistent with a coalescence. If χ_r^2 is greater than unity, $\rho(t)$ is reweighted as $\hat{\rho} = \rho/\{[1 + (\chi_r^2)^3]/2\}^{1/6}$ [29,30]. The final step enforces coincidence between detectors by selecting event pairs that occur within a 15-ms window and come from the same template. The 15-ms window is determined by the 10-ms intersite propagation time plus 5 ms for uncertainty in arrival time of weak signals. We rank coincident events based on the quadrature sum $\hat{\rho}_c$ of the $\hat{\rho}$ from both detectors [10].

To produce background data for this search the SNR maxima of one detector are time shifted and a new set of coincident events is computed. Repeating this procedure $\sim 10^7$ times produces a noise background analysis time equivalent to 608 000 years.

To account for the search background noise varying across the target signal space, candidate and background events are divided into three search classes based on template length. The right panel of Fig. 4 shows the background for the search class of GW150914. The GW150914 detection-statistic value of $\hat{\rho}_c = 23.6$ is larger than any background event, so only an upper bound can be placed on its false alarm rate. Across the three search classes this bound is 1 in 203 000 years. This translates to a false alarm probability $< 2 \times 10^{-7}$, corresponding to 5.1σ.

When an event is confidently identified as a real gravitational-wave signal, as for GW150914, the background used to determine the significance of other events is reestimated without the contribution of this event. This is the background distribution shown as a purple line in the right panel of Fig. 4. Based on this, the second most significant event has a false alarm rate of 1 per 2.3 years and corresponding Poissonian false alarm probability of 0.02. Waveform analysis of this event indicates that if it is astrophysical in origin it is also a binary black hole merger [9].

V. Source discussion

The matched-filter search is optimized for detecting signals, but it provides only approximate estimates of the source parameters. To refine them we use general relativity-based models [23,24,31,32], some of which include spin precession, and for each model perform a coherent Bayesian analysis to derive posterior distributions of the source parameters [33]. The initial and final masses, final spin, distance, and redshift of the source are shown in Table I. The spin of the primary black hole is constrained to be < 0.7 (90% credible interval) indicating it is not maximally spinning, while the spin of the secondary is only weakly constrained. These source parameters are discussed in detail in [11]. The parameter uncertainties include statistical errors and systematic errors from averaging the results of different waveform models.

Using the fits to numerical simulations of binary black hole mergers in [34,35], we provide estimates of the mass and spin of the final black hole, the total energy radiated in gravitational waves, and the peak gravitational-wave luminosity [11]. The estimated total energy radiated in gravitational waves is $3.0^{+0.5}_{-0.5}\, M_\odot c^2$. The system reached a peak gravita-

TABLE I. Source parameters for GW150914. We report median values with 90% credible intervals that include statistical errors, and systematic errors from averaging the results of different waveform models. Masses are given in the source frame; to convert to the detector frame multiply by $(1+z)$. The source redshift assumes standard cosmology.

Primary black hole mass	$36^{+5}_{-4}\,M_\odot$
Secondary black hole mass	$29^{+4}_{-4}\,M_\odot$
Final black hole mass	$62^{+4}_{-4}\,M_\odot$
Final black hole spin	$0.67^{+0.05}_{-0.07}$
Luminosity distance	410^{+160}_{-180} Mpc
Source redshift z	$0.09^{+0.03}_{-0.04}$

Values as published. The citations in the caption are omitted.

tional-wave luminosity of $3.6^{+0.5}_{-0.4}\times10^{56}$ erg/s, equivalent to $200^{+30}_{-20}\,M_\odot c^2/$s.

GW150914 demonstrates the existence of stellar-mass black holes more massive than $\simeq 25\,M_\odot$, and establishes that binary black holes can form in nature and merge within a Hubble time. Binary black holes have been predicted to form both in isolated binaries [36–38] and in dense environments by dynamical interactions [39–41]. The formation of such massive black holes from stellar evolution requires weak massive-star winds, which are possible in stellar environments with metallicity lower than $\simeq 1/2$ the solar value [42,43].

VI. Conclusion

The LIGO detectors have observed gravitational waves from the merger of two stellar-mass black holes. The detected waveform matches the predictions of general relativity for the inspiral and merger of a pair of black holes and the ringdown of the resulting single black hole. These observations demonstrate the existence of binary stellar-mass black hole systems. This is the first direct detection of gravitational waves and the first observation of a binary black hole merger.

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